MD OMAR FARUK MARUF

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Overview

Detail-oriented, organized, fast-paced software Engineer with hands-on experience in blockchain architecture, systems programming, security, fintech, and full-stack web development, holds a B.Sc. in Computer Science & Engineering with an A+ thesis in applied multiprocess computer vision data pipelines, and an IELTS Academic band score of 6.5. At SMC Labs BD, I work on Bitcoin Core's open-source codebase to build TOI Chain, a Layer-1 blockchain, where I designed and developed a crypto wallet and secure communication protocol for nodes. I also build full-stack applications like Agaaw and work with robotics projects like SYNERGY and ERD. I bring strong skills in research & development , fintech business, software and network infrastructure and debugging large codebases.

EDUCATION

University of Global Village (UGV) — Barishal, Bangladesh
Bachelor of Science in Computer Science and Engineering (Feb 2020 – Feb 2024)

- Completed BSc with an A+ graded thesis and projects,
- participated in coding competitions and extracurricular activities and Member of UGV Programing Club
- Advanced and practical courses on Advance Networking and Python by IQAC, UGV

CGPA: 3.69 / 4.00 | **Last Semester CGPA:** 3.80 / 4.00

PUBLICATIONS

Maruf, MD Omar Faruk, Md Riadul Islam, Md Jubayar Alam Rafi. "Non-permissible Mobile Detection to Enhance Security in Bangladeshi Museum: A Multiprocess YOLO-Based Approach." In *6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, pp. 37-42. IEEE, 2024.
DOI: 10.1109/ICEEICT62016.2024.10534325

RESEARCH & PROFESSIONAL EXPERIENCE

Software Engineer | TOI Labs (June 2025 – Present)

- Lead research and development on blockchain core architecture design, architect and developed crypto wallet, and implement custom cryptographic and security features using advanced C++.
- Conducted rigorous open-source codebase analysis (Bitcoin) to develop next-generation blockchain infrastructure, and taking the go-to market strategy in the fintech field and cryptocurrency.
- Developed launchpad for TOI, TOI-Access and multiple websites for business and funding purpose
- Collaborate with Professor Justin Y. Shi, from Temple University, USA for research and development of TOIChain: High Performance Tamper Resistant Transactions System without Scaling Limits.
- Field of expertise: Bitcoin, C++, Blockchain, dApp, NextJS, NestJS, JavaScript, Crypto Business

Software Intern (Research & Development) | SMC Labs BD (Dec 2024 – May 2025)

- Collaborated with senior software engineers on open-source Bitcoin core research, decentralization technologies, and system performance tuning.
- Authored detailed technical documentation and research specifications for enterprise dApp integration.

Python Trainer | Sheikh Russel School of Future, ICT Division (May 2024 – Oct 2024)

- Instructed secondary school students (Classes 6–10) in foundational computer science, algorithm development, and data structures using Python.
- Prepared and mentored students for national robotics competitions and government-funded tech initiatives.

PROJECTS

AGAAW | Peer-to-Peer Academic Mentorship Platform (Startup Project)

- Designed, developed and deployed the whole platform, both frontend and backend, leading the team members as a CTO role.
- Integrated real-time video conferencing, dynamic chat architecture, scholarship search algorithms, and geographical discovery tools.
- Helping to take business initiatives, and go-to market strategy through connecting with Venture capitals and investors.
- Field of Expertise: Leadership, NextJS, NestJS, JavaScript,, HTML & CSS, WebSocket, APIs, AI

HONORS & AWARDS

- **Champion – GP Accelerator (Grameenphone Ltd, 2024):** Developed the Emergency Response Device (ERD) IoT system for automated environmental alerting, winning 100,000 BDT prize money among 17 finalist teams.
- **Champion – imaGen Ventures Youth Challenge (UNICEF Bangladesh, 2023):** Developed the decision-making and multi-process control software for "SYNERGY" autonomous plastic-collection robot, winning $1,000 prize money among 200 competing teams.

TECHNICAL SKILLS

- **Languages:** C++ (Advanced), Python, Solidity, JavaScript & TypeScript
- **Core Domains:** Decentralized Application, Blockchain Infrastructure, Real-time Systems, Fintech, Database Management (PostgreSQL, Computer Vision, Deep Learning, Web-Development.
- **Frameworks & Tools:** NextJS, NestJS, Git/GitHub, Docker, Redis, PostgreSQL, Linux Environments, Google workspace
- **Soft Skills:** Teamwork, Leadership, Communication, Project management, Problem solving

LEADERSHIP & EXTRACT-CURRICULAR ACTIVITIES

President | DYDF, Babuganj Branch (Nov 2024 – Present)

- Lead district-level youth empowerment, educational literacy programs, and community technology awareness campaigns.

Active Volunteer Member | Volunteer for Bangladesh (VBD) (Oct 2023 – Oct 2024)

- Organized emergency disaster response and relief operations for flood-affected populations.

CERTIFICATIONS & STANDARDIZED TESTS

- **IELTS Academic:** Overall Band 6.5 (Speaking: 6.5, Listening: 6.5, Writing: 6.5, Reading: 6.0)
- **Coursework:** Introduction to Cybersecurity, CISCO | AI Intensive Course, Google X Kaggle | Advance Networking, UGV | C++, Python, DSA, SQL (SoloLearn)
- **Languages:** Bengali (Native), English (Fluent), German (A1 Basics)